STRENGTH OF MATERIALS AND MATERIAL SCIENCE AND METALLURGY LAB

II B TECH I Semester : MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6ME11	PCC	0	0	2	1	40	60	100

COURSE OUTCOMES:

At the end of the course, students will be able to

1. Analyze mechanical behaviour of test sample under tensile, compressive, torsion and impact load conditions.
2. Find deflection of simply supported and cantilever beams with different cross sections/materials.
3. Calculate stiffness and modulus of rigidity of helical coil spring.
4. Identify the various phases depicted in the microstructure of pure metals, ferrous and non-ferrous alloys.
5. Analyze the significance of cooling rate in the formation of fine grains using Jominy end quench test.

SYLLABUS:

LIST OF EXPERIMENTS: Strength of Materials Lab

1. Direct tension test.
2. Deflection test on simply supported beam and Cantilever beam.
3. Torsion test.
4. Hardness test
   a. Brinell's hardness test
   b. Rockwell hardness test
5. Test on springs.
6. Charpy and Izod Impact test.

LIST OF EXPERIMENTS: Metallurgy Lab

7. Study of microstructures of pure metals such as Iron, Cu, Al, etc.
8. Study of microstructures of plain carbon steels.
9. Study of microstructure of cast irons.
10. Study of microstructure of alloy steels.
11. Study of microstructures of non-ferrous alloys.
12. Determination of hardenability of steels by Jominy end quench test.